



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	HW 4 Laplace & Fourier Transform	Marks 60 Weight 2.5%	Due: 26 th April, 2026 (online submission).
-------------	-----------	--	-------------------------	---

Important Notes

Solution should be elaborative, sketch neat time & frequency domain plots wherever required, show clearly all steps of calculation & mention reference (for example, formula, law, rule, transform property) to strengthen your understanding to solve the SnS related problems.

Also, please comply with the Academic Integrity Policies of Habib University related to plagiarism while submitting this assignment for grading.

Grading Rubric:

Steps of working and logical flow of solution: 80%

Final answer value of the given question: 20%

Late Submission Policy

As announced in the course syllabus.

CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
--------------	--	--------------------------------	-------------------------------------

Problem 1 [10 pts]

Find the inverse Laplace transforms of the following functions:

(a) $\frac{(2s + 5)e^{-2s}}{s^2 + 5s + 6}$

(b) $\frac{se^{-3s} + 2}{s^2 + 2s + 2}$

Problem 2 [10 pts]

Consider a causal LTIC system described by the differential equation

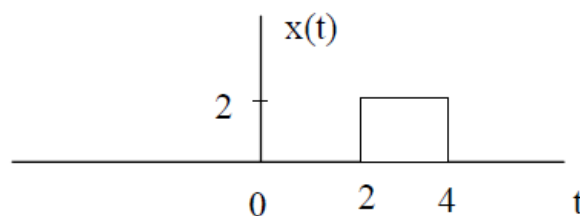
$$2\dot{y}(t) + 6y(t) = \dot{x}(t) - 4x(t)$$

- (a) Using transform-domain techniques, determine the ZIR $y_{zir}(t)$ if $y(0^-) = -3$.
- (b) Using transform-domain techniques, determine the ZSR $y_{zsr}(t)$ to the input $x(t) = e\delta(t - \pi)$.

Problem 3 [10 pts]

Find the Fourier transform of the following signals. Sketch $|X(\omega)|$ in each case.

a)



b) $x(t) = 2e^{-2t}u(t)$

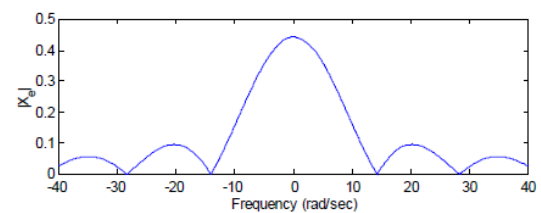
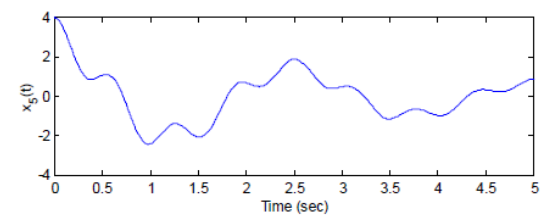
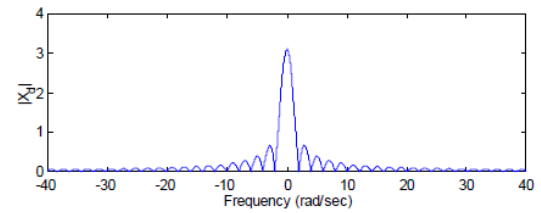
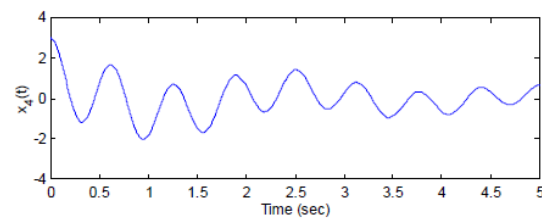
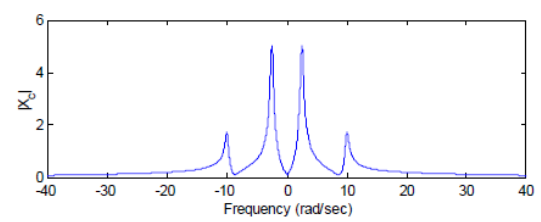
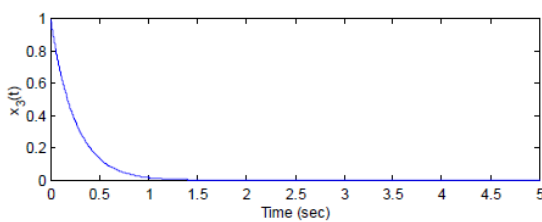
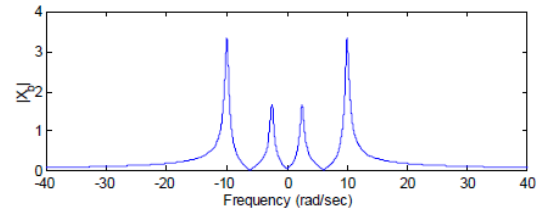
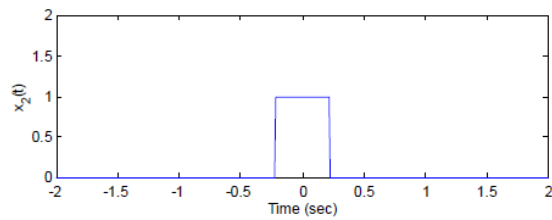
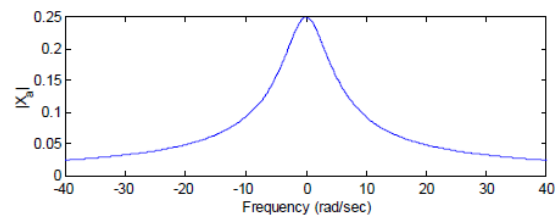
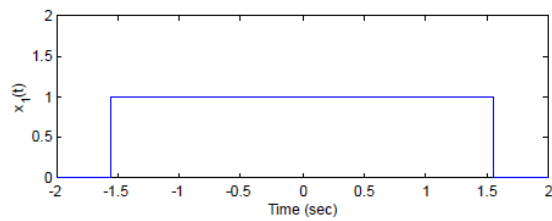
c) $x(t) = 5e^{-5t}u(t)$

d) $x(t) = e^{-2t}\cos(4t)u(t)$

Problem 4 [10 pts]

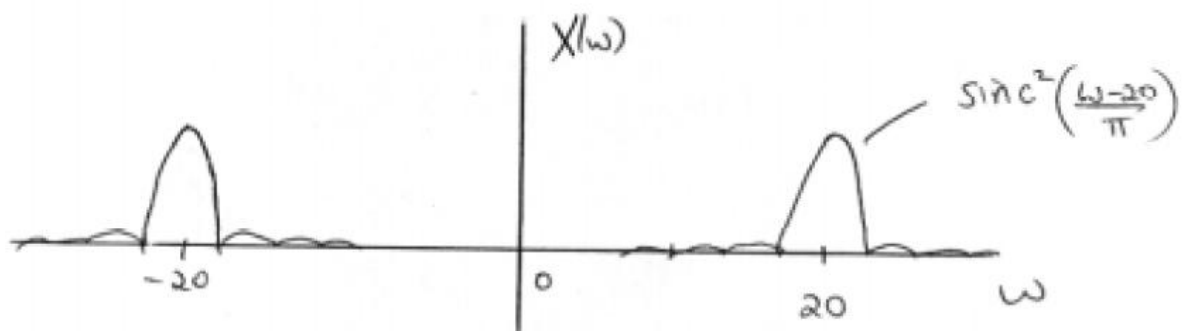
Match the time responses with the corresponding frequency responses.

Justify your choices by providing a brief description.



Problem 5 [10 pts]

Compute the inverse Fourier transform of the following signal.



Problem 6 [10 pts]

An **Exponential Fourier Series** is defined as

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

Whereas, a **Trigonometric Fourier Series** is defined as

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos k\omega_0 t + b_k \sin k\omega_0 t)$$

Find the relationship between the coefficients (c_k, a_0, a_k, b_k) of the two series.